

LESSON 7-7 • SESSION 2 • PRACTICE • FORM B**Designing Polygons with Scale**

Learning target – I can use a scale to find a missing dimension of a figure on the coordinate plane. · 6.G.A.3

REMEMBER

- A **scale** tells how long one grid unit is in real life: 1 unit = 4 m.
- Change each side to real length **first** (units $\times$ scale). Then find perimeter or area.
- Missing side = **area $\div$ known side**. Area uses square units (ft^2 , m^2).

WORKED EXAMPLE

A plan uses 1 unit = 10 m. A field must cover 2,400 m^2 and is 4 units wide. Find its real length.

- Step 1** Real width: $4 \times 10 = 40$ m.
Step 2 Real length: $2,400 \div 40 = 60$ m.
Step 3 On the plan: $60 \div 10 = 6$ units long.
Answer 60 m long (6 units on the plan)

WORDS scale (escala) · area (área) · dimensions (dimensiones)**PRACTICE**

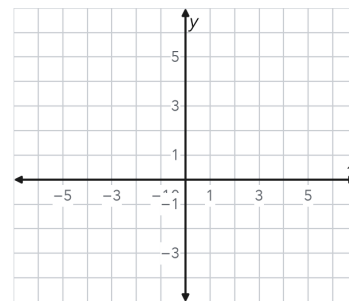
show your work · label units

1. On a plan, 1 grid unit = 8 ft. Change each measure.

- a. 5 units = ____ ft b. 7 units = ____ ft
 c. 48 ft = ____ units d. 96 ft = ____ units

2. **Courtyard.** Three corners of a rectangular courtyard are $(-6, 5)$, $(6, 5)$ and $(6, -4)$. Each unit is 10 feet.

- a. Plot the corners. Name the fourth corner. _____
 b. Width and height in feet _____
 c. Area _____ ft^2



3. **Parking Pad.** Parking pad corners: $(-6, 4)$, $(6, 4)$, $(6, -5)$ and $(-6, -5)$. Each unit is 2 feet. Find the area in square feet.

Hint Change both sides to feet before you multiply.

Show your work.

Answer _____ ft^2

4. **Using the Scale.** On a plan, 1 unit = 20 feet. A rectangle is 5 units by 3 units. Find the real perimeter.

Show your work.

Answer _____ ft

5. **Missing Dimension.** A rectangular lot must cover 4,200 ft^2 . One side is 70 ft. Find the other side.

Hint Divide the area by the known side.

Show your work.

Answer _____ ft

- 6. Fence Cost.** Yard corners: $(-5, 4)$, $(7, 4)$, $(7, -3)$ and $(-5, -3)$. Each unit is 3 feet. Fencing costs \$8 per foot. Find the fence cost.

Hint Change both sides to feet, add all four, then multiply by 8.

Show your work.

Cost _____

- 7. Pen Design.** Kofi wants a pen with an area of 360 ft^2 . One side runs from $(-3, 5)$ to $(6, 5)$. Each unit is 2 feet. Find the other side.

Hint Change the known side to feet first.

Show your work.

Answer _____ ft

- 8. Playground Design.** A plan uses 1 unit = 10 feet. A playground must cover $4,800 \text{ ft}^2$. On the plan it is 6 units wide.

- a. Real width _____ ft
- b. Real length _____ ft
- c. Length on the plan _____ units
- d. Real perimeter _____ ft

FIND THE ERROR

explain the mistake, then fix it

- 9. Kenji's Work.** A side runs from $(-7, 3)$ to $(5, 3)$. Each unit is 8 feet. Kenji wrote the side length as 12 feet. Explain his mistake. Then find the correct length.

TEST PRACTICE

choose the best answer

- 10.** An area of 900 m^2 with one side of 30 m has another side of –
A 870 m **B** 30 m **C** 27,000 m **D** 60 m
- 11.** If 1 unit represents 15 meters, then 6 units represent –
A 21 meters **B** 9 meters **C** 90 meters **D** 2.5 meters

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

\$912 O	108 Q	800 X	16 J	60 Y	80 T	20 U	320 A	\$114 Z	432 L
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#3	#4	#5	#6	#7	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____